SIMATS ENGINEERING

Department of Computer Science and Engineering

CSA15 Cloud Computing and Big Data Analytics

CO2 AT3 - Capstone Cloud Infrastructure Case Analysis and Defense

Editable student template for application-based cloud virtualization and SDDC/cloud operations defense

Assessment Metadata

Course Code	CSA1517
Course Title	Cloud Computing and Big Data Analytics
Assessment	CO2 AT3 - Capstone Cloud Infrastructure Case Analysis and Defense
Submission Type	Individual digital case analysis and infrastructure defense based on the same capstone title used in CO1, CO2 AT1 and CO2 AT2
Faculty	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Submission Mode	Completed DOCX template and GitHub repository link through Google Classroom

How This Template Works

- Use the same capstone title used in CO1, CO2 AT1 and CO2 AT2.
- Do not recalculate all VM sizing or repeat all configuration screenshots. Summarize the final AT1/AT2 evidence and use it for design defense.
- Answer all five questions with application-oriented reasoning connected to the capstone product.
- Include sustainable cloud operation decisions wherever relevant: right-sized resources, log retention, storage lifecycle, scheduled shutdown, API-call reduction and VM/container/serverless comparison.
- Paste the final topology diagram in the given response area. The sample diagram is only a structure; redraw it for the student's own product.
- Upload this completed document and supporting evidence to GitHub, then submit the repository link through Google Classroom.

Assessment Focus

AT1 answered what VM is required. AT2 showed configuration or simulation evidence. AT3 asks whether the student can defend the complete infrastructure design under application, security, operations and failure conditions.

Student and Capstone Details

Student Name	DENVAR JOE H
Register Number	192511438
Class / Slot	Slot C

Capstone Title	HostelMate Green Cloud: Sustainable Hostel Management with Energy-Efficient SLA Optimization and Geo-Tagged Complaint Intelligence
CO1 Evidence Link	
CO2 AT1 VM Recommendation	vCPU: 2 RAM: 4 GB Storage: 50 GB VM Class: General Purpose (B2s)
CO2 AT2 Evidence Link	
GitHub Repository Link	https://github.com/Denvarj/CLOUD-COMPUTING-AND-BIG-DATA.git
Submission Date	04-08-2026

Evidence Carry-Forward Summary

Evidence Source	Student Entry	How It Supports AT3 Defense
CO1 concept map / presentation	Problem statement, SDG 11 & SDG 12 objectives, system modules and cloud architecture	Demonstrates the project objectives, cloud-based design, and sustainability goals.
CO2 AT1 sizing workbook	VM recommendation: 2 vCPU, 4 GB RAM, 50 GB Storage, General Purpose VM	Justifies the selected infrastructure size and resource allocation for the application.
VM calibration sheet	Performance estimation and resource utilization analysis	Confirms that the selected resources are adequate for the expected workload.
CO2 AT2 practical evidence	Firebase setup, Cloudinary integration, Flutter application, security configuration, and GitHub repository	Demonstrates successful implementation, deployment readiness, and secure cloud configuration.
Capstone architecture decision	Firebase Authentication, Firestore, Firebase Storage, Cloud Functions, Cloudinary, Google Maps API, React Dashboard	Supports the final cloud architecture, scalability, automation, security, and Green Cloud objectives.

Q1: Virtualization Value and Capstone Cloud Deployment Summary

Explain your capstone product, expected workload, cloud need and deployment context. Analyze how virtualization improves cost efficiency, resource utilization, workload isolation, infrastructure flexibility and cloud deployment readiness for your product.

Design Point	Student Response
Capstone product and users	HostelMate Green Cloud is a cloud-based hostel complaint management system used by students, wardens, maintenance staff, vendors, and administrators to report, manage, and monitor hostel maintenance issues efficiently.
Expected workload	The application handles daily complaint submissions, image uploads, location tagging, notification delivery, analytics processing, and multiple users accessing the system simultaneously.
Cloud need	Cloud infrastructure provides scalable storage, secure authentication, real-time database synchronization, serverless processing, and anywhere access without maintaining physical servers.
Virtualization value	Virtualization allows cloud providers to efficiently share physical hardware among multiple services, improving resource utilization, scalability, isolation, and system reliability.
Cost and utilization benefit	Instead of purchasing dedicated servers, the project uses Firebase managed cloud services that automatically allocate resources based on demand, reducing operational cost and minimizing idle resources.
Isolation and flexibility benefit	Each cloud service such as Authentication, Firestore, Storage, Cloud Functions, and Cloudinary operates independently. This modular architecture simplifies maintenance, improves security, and allows future feature expansion.

Case Analysis Response: Write 8-12 technical lines connecting virtualization value to your capstone product.

HostelMate Green Cloud is developed to automate hostel maintenance management through cloud computing. Students can submit complaints with photos and GPS locations using the Android application. Firebase provides authentication, real-time database services, cloud storage, and serverless backend processing. Virtualization enables efficient allocation of computing resources without requiring dedicated physical servers. Cloud Functions automatically process complaint routing and SLA monitoring. Firestore synchronizes complaint data across all users in real time. Cloudinary stores and optimizes uploaded images, reducing storage overhead and improving delivery performance. Managed cloud services increase reliability, scalability, and fault tolerance while lowering infrastructure costs. The modular cloud architecture also simplifies maintenance and future upgrades. Overall, virtualization makes the application flexible, energy efficient, and suitable for sustainable cloud deployment.

Q2: Rapid Provisioning, Testing and Virtual Machine Lifecycle Case

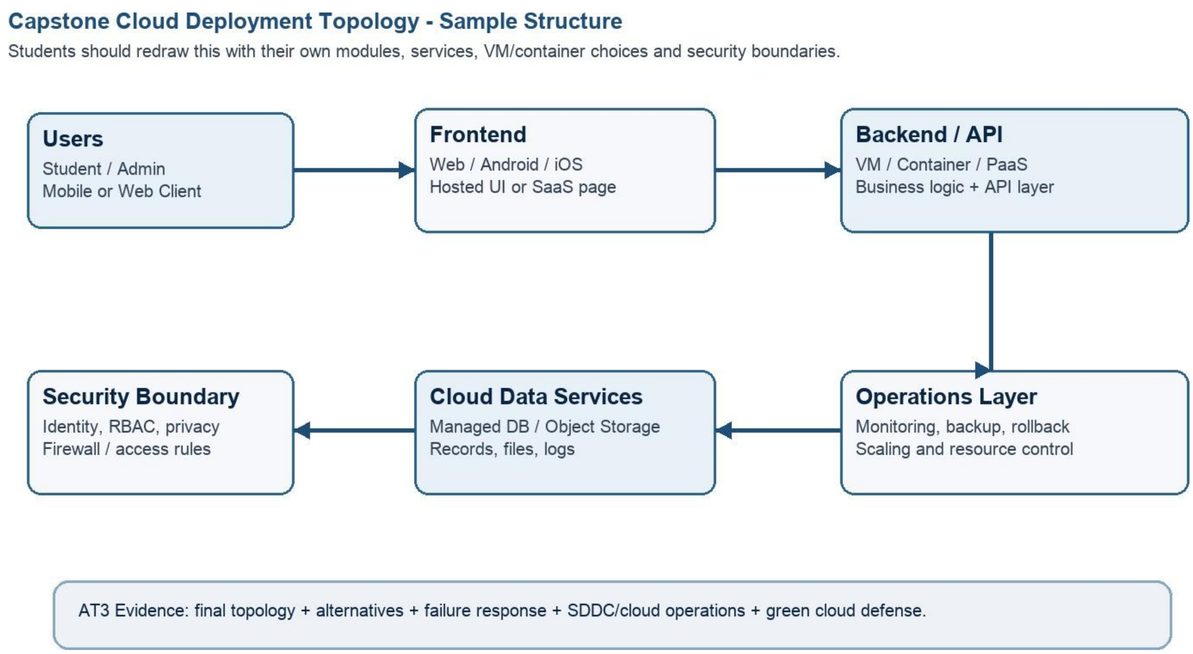
Using your CO2 AT1 VM recommendation and CO2 AT2 practical evidence, explain how virtual machines, VM images, templates, snapshots, clones or containers support rapid provisioning, testing, deployment, rollback and maintenance for your capstone product.

Lifecycle Element	AT1/AT2 Evidence Used	Capstone Use / Justification
VM image or base environment	Firebase Project + Flutter SDK	A standard Flutter and Firebase development environment is used for all developers to maintain consistency.
Template or repeatable setup	Firebase project configuration and GitHub repository	New developers can clone the repository and configure Firebase quickly using the existing project setup.
Snapshot / rollback	GitHub Version Control and Firebase Backup	Previous application versions can be restored if deployment introduces errors or failures.
Clone / testing copy	GitHub Branches and Firebase Test Project	Separate testing copies are created before deploying new features to production.
Container or VM decision	Firebase Console and Cloud Functions	Updates are deployed with minimal downtime while monitoring performance through Firebase Console.
Maintenance and update plan		

Case Analysis HostelMate Green Cloud uses managed cloud services to simplify application deployment and maintenance. The Flutter application is developed using a standardized Firebase environment, allowing quick setup for every developer. GitHub stores the complete source code, enabling version control, branching, and rollback whenever issues occur. A separate Firebase testing project is used to validate new features before production deployment. Since Firebase provides serverless backend services, virtual machine management is not required, reducing operational complexity and maintenance costs. Cloud Functions automatically execute backend tasks such as complaint routing and SLA monitoring without manual server configuration. Cloudinary securely stores uploaded complaint images while optimizing delivery performance. This lifecycle approach reduces deployment time, minimizes downtime, improves reliability, and supports continuous development with lower infrastructure risk.Response: Explain how lifecycle practices reduce deployment delay and risk.

Q3: Final Cloud Deployment Topology and SOA/API Service Integration Case

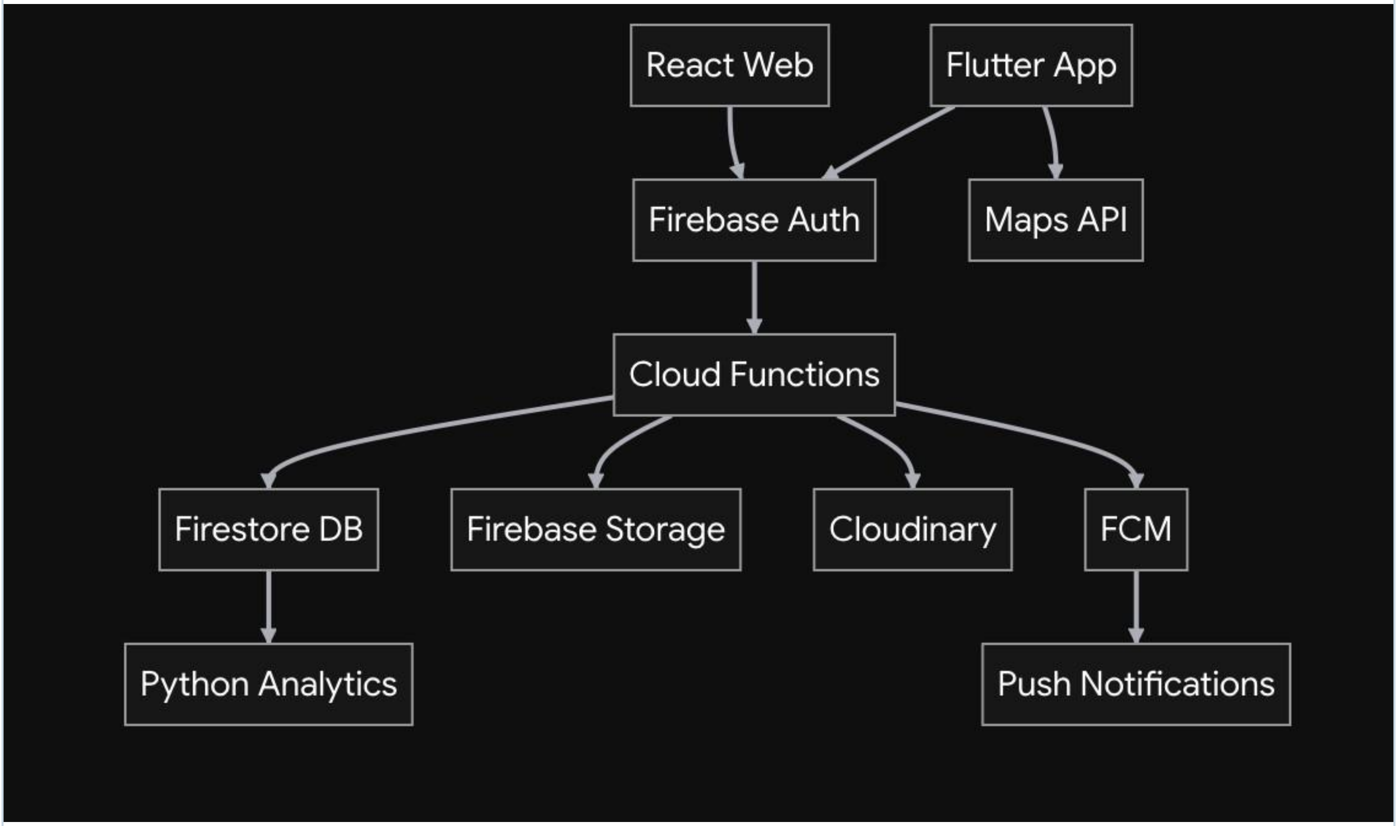
Draw and explain the final deployment topology for your capstone product. Show frontend, backend/API, database, storage, authentication, notification, analytics and any VM/container/managed cloud service components. Explain how SOA/API-based integration connects these modules and supports modular cloud service design.



Architecture Component	Technology / Service	VM / Container / Managed Service / SaaS	Integration Method
Frontend	Flutter Android App	Client Application	Uses Firebase SDK and REST APIs
Backend/API	Firebase Cloud Functions	Managed Serverless Service	Triggered by Firestore events and HTTPS APIs
Database	Firebase Firestore	Managed NoSQL Database	Real-time synchronization with Flutter
Storage	Firebase Storage + Cloudinary	Managed Cloud Storage	Stores complaint images and retrieves image URLs
Authentication	Firebase Authentication	Managed Authentication Service	Email/Password login with Role-Based Access Control
Notification	Firebase Cloud Messaging (FCM)	Managed Notification Service	Sends complaint status and SLA notifications

Analytics / AI module	Python Analytics	Managed/Cloud Hosted	Reads Firestore data for SLA reports, priority analysis, and complaint trends

Paste or draw the final capstone deployment topology diagram here.



SOA/API Integration Explanation: Explain how services communicate and why modular service design is useful.

HostelMate Green Cloud follows a Service-Oriented Architecture (SOA) using multiple independent cloud services. The Flutter application communicates with Firebase Authentication for secure user login and Firestore for storing complaint information. Images uploaded by users are stored in Firebase Storage and Cloudinary, while Google Maps API provides location tagging for each complaint. Firebase Cloud Functions automate ticket assignment, SLA monitoring, and workflow processing without requiring dedicated servers. Firebase Cloud Messaging sends real-time notifications to students and maintenance staff whenever complaint statuses change. The Python Analytics module processes complaint data to generate reports, identify recurring infrastructure issues, and evaluate vendor performance. The React web dashboard displays analytics and complaint management information for administrators. This modular architecture improves scalability, simplifies maintenance, enhances security, and allows each service to be updated independently without affecting the entire application.

Q4: Design Alternatives, Identity, Access and Federated Cloud Privacy Case

Compare at least two deployment alternatives for your capstone product, such as single VM, multi-tier VM, containerized deployment, managed backend, federated cloud or hybrid cloud model. Justify the selected design by analyzing identity management, role-based access control, federated access, data sharing, privacy protection and security risks.

Deployment Alternative	Advantages	Limitations / Risks	Identity and Privacy Impact
Alternative 1	Easy to deploy and manage for small applications.	Limited scalability, higher maintenance, manual updates, and single point of failure.	User authentication and data security depend entirely on the VM configuration, increasing security risks.
Alternative 2	Automatic scaling, serverless architecture, high availability, built-in authentication, lower maintenance, and better reliability.	Vendor lock-in and dependency on internet connectivity.	Secure authentication, encrypted data transmission, and role-based access control improve user privacy and security.
Selected design	Firebase Managed Cloud with Cloudinary	Best balance of scalability, security, cost, and operational efficiency.	Provides secure identity management and protects sensitive hostel complaint data.

Security Area	Student Decision
User roles	Student, Warden, Maintenance Staff, Vendor, Administrator
Authentication method	Firebase Authentication using Email and Password
Role-based access control	Firestore Security Rules restrict users based on their assigned roles. Students can access only their complaints, while administrators and staff have additional permissions.
Federated access / partner access	Vendors receive access only to maintenance tickets assigned to them. They cannot view unrelated hostel data.
Sensitive data handled	Student details, complaint descriptions, hostel locations, uploaded images, maintenance history, and SLA records.
Privacy protection decision	Firebase Authentication secures user identity, Firestore Security Rules prevent unauthorized access, HTTPS encrypts data in transit, and Cloudinary securely stores complaint images.

Case Analysis Response: Defend why the selected design is better for identity, access and privacy.

HostelMate Green Cloud considered both a traditional virtual machine deployment and a managed cloud backend architecture. A single virtual machine requires manual server maintenance, software updates, resource scaling, and security management, making it less suitable for a growing application. The selected design uses Firebase managed cloud services with Cloudinary, which automatically provide scalability, availability, and serverless infrastructure management. Firebase Authentication verifies user identities, while Firestore Security Rules enforce role-based access control for students, wardens, maintenance staff, vendors, and administrators. Vendors can access only the maintenance tasks assigned to them, protecting hostel data from unauthorized access. Sensitive information, including complaint details, locations, and uploaded images, is securely stored using encrypted cloud services. This architecture improves privacy, reduces operational complexity, lowers maintenance costs, and provides a secure and scalable platform for sustainable hostel management.

Q5: Software-Defined Datacenter and Cloud Operations Defense Case

Analyze how your infrastructure will respond to operational and failure scenarios such as VM failure, database growth, traffic spike, credential leakage, service outage, resource contention or cost increase. Explain how SDDC concepts and cloud operations practices such as automated provisioning, software-defined networking, software-defined storage, monitoring, logging, backup, rollback, scaling, resource control and maintenance improve the final infrastructure design. Include Green Computing decisions such as resource right-sizing, storage lifecycle control, log retention, API-call reduction, scheduled shutdown, VM/container/serverless comparison and carbon/energy proxy indicators where relevant.

Operational / Failure Scenario	Expected Impact	SDDC / Cloud Operations Response	Evidence or Control
VM failure	Backend services become unavailable.	Firebase is a managed serverless platform, so there is no single VM failure. Google Cloud automatically handles infrastructure availability.	Firebase managed infrastructure
Database growth	Slower query performance and increased storage usage.	Firestore automatically scales storage and performance as data grows.	Firestore Auto Scaling
Traffic spike	Large number of users may access the application simultaneously.	Firebase automatically scales backend services to handle increased traffic.	Auto Scaling
Credential leakage	Unauthorized access to user accounts.	Firebase Authentication, password reset, Firestore Security Rules, and role-based permissions protect accounts.	Authentication & RBAC

Service outage	Users may temporarily lose access to services.	Cloud monitoring, retries, and backup services minimize downtime.	Firebase Status & Backup
Resource contention	Performance degradation during high usage.	Serverless architecture automatically allocates resources as needed.	Dynamic Resource Allocation
Cost increase	Higher cloud billing due to increased usage.	Monitor usage, optimize storage, compress images, and remove unused data.	Firebase Usage Dashboard

Operational Readiness Area	Student Plan
Automated provisioning	Firebase automatically provisions backend resources without manual server configuration.
Software-defined networking	Google Cloud securely manages network communication between Firebase services.
Software-defined storage	Firestore, Firebase Storage, and Cloudinary automatically manage scalable cloud storage.
Monitoring and logging	Firebase Console and Cloud Logging are used to monitor application performance and errors.
Backup and rollback	GitHub version control and Firestore database backups allow quick recovery from failures.
Scaling and elasticity	Firebase automatically scales according to user demand without manual intervention.
Resource control and maintenance	Remove unused data, optimize images, and regularly update Firebase security rules.
Green Computing and sustainability control	Use serverless architecture, automatic scaling, optimized storage, and reduced energy consumption.

Green Cloud Decision	Student Justification
VM right-sizing from CO2 AT1	Firebase serverless services eliminate the need to maintain oversized virtual machines, reducing unnecessary resource usage.
VM vs container vs serverless choice	Serverless is selected because Firebase automatically manages infrastructure, scaling, and maintenance while reducing operational overhead.
Scheduled shutdown / idle-resource control	Serverless resources are allocated only when requests are processed, minimizing idle resource consumption.
Log and file retention policy	Old logs and completed complaint records are archived or deleted periodically to reduce storage usage.

Storage lifecycle and backup cleanup	Complaint images that are no longer required are removed after a defined retention period while backups are maintained for important records.
API-call or polling reduction	Firestore real-time listeners are used instead of continuous polling, reducing unnecessary API requests and network traffic.
Carbon/energy proxy indicator to track	Monitor cloud storage usage, number of Cloud Function executions, database reads/writes, and estimated energy savings through reduced idle infrastructure.

Final Infrastructure Defense Statement: State whether the design is ready for prototype, pilot or production. Defend reliability, security, SDDC/cloud operations and Green Computing improvements required next.

The proposed infrastructure is **ready for prototype and pilot deployment**. Firebase managed cloud services provide automatic scaling, secure authentication, high availability, and serverless backend processing, while Cloudinary efficiently stores and delivers complaint images. Firestore enables real-time synchronization, and Cloud Functions automate ticket routing and SLA monitoring. Security is maintained through role-based access control, Firestore Security Rules, encrypted communication, and authenticated user access. Green Computing principles are implemented using serverless architecture, automatic resource scaling, optimized storage, reduced API calls, and data retention policies. Future improvements include advanced AI-based complaint prediction, automated vendor performance analysis, and enhanced monitoring dashboards to support full-scale production deployment.

CO2 Reflection and Individual Defense

This section must be individually written. It should show what the student understands from the complete CO2 chain: infrastructure sizing, practical configuration evidence and final operations defense.

Reflection Prompt	Student Response
What is the strongest infrastructure decision in your capstone design?	Choosing Firebase serverless architecture because it provides automatic scaling, security, and low maintenance without managing virtual machines.
Which CO2 concept became clearer after AT1, AT2 and AT3?	The relationship between virtualization, cloud deployment, managed services, and scalable infrastructure became much clearer.
Which risk is most serious for your capstone: failure, cost, privacy, scaling or sustainability?	Privacy is the most critical risk because the system stores student information, complaint details, locations, and uploaded images.
What evidence would you show during viva to defend your infrastructure design?	Firebase project configuration, Firestore database, Cloud Functions, Flutter application, Cloudinary integration, deployment topology, and GitHub repository.
What will you improve before final capstone deployment?	Add AI-based complaint prediction, stronger monitoring, automated backups, vendor analytics, and improved security auditing.

Student-Facing Assessment Criteria

Criteria	Descriptor
Capstone continuity	The answer uses the same capstone title and connects CO1, CO2 AT1 and CO2 AT2 evidence.
Virtualization and lifecycle reasoning	VM, image, snapshot, clone, container or managed-service decisions are technically justified.
Topology and integration clarity	Architecture diagram and SOA/API explanation show clear module interaction.
Identity, access and privacy	Roles, authentication, RBAC, federated access, sensitive data and privacy decisions are defended.
SDDC/cloud operations	Failure response, monitoring, logging, scaling, backup, rollback and resource control are addressed.
Green Computing defense	Right-sizing, retention, API efficiency, idle-resource control and sustainable cloud choices are explained.
Submission quality	GitHub, template completion, evidence references and individual reflection are complete.

GitHub and Google Classroom Submission

Repository Item	Expected Content
README.md	Capstone title, student details, final VM recommendation, deployment topology summary and AT3 evidence index.
Completed AT3 document	This completed editable DOCX or exported PDF if instructed.
Diagram image	Final topology diagram if created separately.
Evidence references	Links or screenshots from CO1, CO2 AT1 and CO2 AT2 as needed.
Green Computing note	Short README section explaining sustainable cloud choices and resource-control decisions.
Google Classroom	Submit the GitHub repository link through GCR.

Student Assurance

I confirm that this case analysis, design reasoning, topology diagram and final defense statement were prepared individually by me for the stated capstone title. I have not uploaded passwords, tokens, private keys or sensitive account information.

Student Name	DENVAR JOE H
Register Number	192511438
Signature	
Date	

Faculty Verification

Faculty Name	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Constructive Feedback / Remarks	
Strength Observed	
Improvement Suggested	
Signature / Date	